

Laura Werthmann Dozal

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EDUCATION

PhD in Information Science, Minor in Statistics and Data Science; Certificate in Computational Social Science,
College of Information, University of Arizona, Tucson, Arizona - May 2026

Outstanding PhD Student Award 2026

Dissertation: Understanding Social Movement Messaging from a Collection of Topical Images: Using Multimodal Sentiment Analysis for Narratives in Structured Graphs.

Master of Arts in Communication, Culture, and Technology; *Georgetown University, Washington, D.C. 2016*

Bachelor of Arts in English Literature; Minor in Communication; *St. Edward's University, Austin, Texas 2012*

Bilingual: Fluent in English and Spanish

PUBLICATIONS

- **"When VR Was New": Looking Back at Undergraduate Students' Initial Encounters with Virtual Reality in an Academic Library.** *The Library Quarterly.* Forthcoming, 2026
 - **Jetstream2 AI Workshop Project Manager.** - February 2026. doi:10.5281/zenodo.18407373 "AI Pipeline for Synthetic Data Generation, Generative M-LLM Comparison, and Network Building"
 - **Visual Semantics of Social Movements Online: Generative Models and Network Analysis for Social Media Messaging,** 2025. *ASIS&T Conference Proceedings - Short Paper Publication, Crystal City VA*
 - **Mixed Methods Framework for Understanding Visual Frames in Social Movements,** October 2023. *ASIS&T Conference, Long Paper Publication, London UK*
 - **Extracting Space Situational Awareness Events from News Text,** May 2022. *LREC Conference. Paper Publication*
 - **Detecting the Spread of Propaganda Bias Online: Comparing HAN and BERT Models,** 2021. *TPRC, Paper*
 - **Data Visualization - Old North Pilgrimage Project,** 2016, *Permanent Art Installation, Georgetown University*
 - **Network Installations - An Art Technology,** 2015. *Paper Publication Presenter, PCA/ACA. STGlobal Conference, DC*
 - **Street Art Dialogue,** Dec. 2014. *Gnovis Journal - Peer Reviewed Publication*
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SPEAKING ENGAGEMENTS and CONFERENCES

- Invited Seminar Speaker: **Women in Network Science (WiNS) Spring 2026** – Synthetic Data for Semantic Analysis
- Invited Speaker: **Jestream2 AI Fellows Orientation January 2026:** *Using Multimodal Sentiment Analysis for Knowledge Retrieval in Structured Graphs: Understanding Narrative Structure from a Collection of Topical Images.*
- Invited Panelist: **Women in Data Science (WiDS) 2025** – Generative AI and the Use of AI in Automation
- **iConference 2024:** Long Paper Submission Reader
- **Conference of Complexity Science,** Sept. 2024, Exeter, England; **Poster:** *Visual and Textual Analysis of the Online Anti-Femicide Movement in Mexico: A Narrative Network Approach*
- **Visual and Textual Analysis of the Online Anti-Femicide Movement in Mexico: A Narrative Network Approach,** *ASONAM2024 IEEE Conference; and ACM GOODIT2024 Conference – Accepted Short Papers, could not attend.*
- **Association for Computational Linguistics (ACL) Conference,** July 2024 - **WOAH workshop presentation;** *Visual and Textual Narrative Analysis of the Online Anti-Femicide Movement in Mexico: A Narrative Network Approach*
- **Winter Workshop on Complex Systems (Santa Fe Institute),** Jan. 2024; **Presenter** and Participant: Contemplated future challenges of complexity science including interdisciplinarity, scalability, causality, AI, and ethics.
- **University of Rochester Libraries,** **Presenter** for LEADING Data Fellows, Dec. 2023; *Bibliometric Data: Publishing Behavior of University of Rochester Researchers*
- **University of Arizona iSchool Colloquium,** Nov. 2023; **Presenter** - *Working with Foundational Open-Source Science Skills (FOSS) for Dissertation Data and Reproducibility*
- **ASIS&T Workshop Panelist:** "Turning Social Informatics Research into Action in a Changing Moment", Oct. 2023, London UK; *Narrative Frames in Topical Images: Mixed Methods Approach to Understanding Social Movements*
- **Posit:Conf2023,** Sept. 2023 – Chicago, IL; *Opportunity Scholar, Registration, travel, room and board paid for.*
- **International Conference on Computational Social Science (IS2C2),** July 2022, Chicago, IL; **Poster,** *Mixed Methods for Understanding Visual Frames in Social Movements*
- **ASIS&T 2021; Poster:** *Street Art as Visual Information: Mixed Methods Approach to Analyzing Community Spaces*

EXPERIENCE & RESEARCH**Instructor of Record** | *University of Arizona College of Information* - Tucson, AZ, 08/2019 – Present

- ISTA 116: Statistical Foundations for the Information Age, **Spring/Fall Semesters 2024-Present**
- ISTA 130 – Computational Thinking and Doing (Introduction to Python), **Summers 2021 - Present**
- ESOC 300 – Digital Storytelling and Culture, Summer 2020

Jetstream2 AI Fellow | NSF Funded Program through University of Indiana; Remote 1/13/2025 – 1/06/2026

- **Compute and Processing Resources Awarded:** Priority access to 90 GPU-enabled nodes with four NVIDIA A100 GPUs processing resources with instance allocation. Jetstream2 AI Workbench and other training for machine learning and Artificial Intelligence (AI/ML) workflows. Iteratively build an AI/ML workbench organized for reproducibility and 1 day workshop training. **Awarded \$5,000 grant**
- Created an LLMops/MLOps pipeline to lead the Open Science project that holds a version control codebase (GitHub, docker, etc) for data collection, computer vision classification models, semantic clustering, and graph database development. Used multimodal generative AI models for image labeling and captions. Comparative analysis of CNNs, vision transformers (ViTs), multimodal(vision-language), and diffusion-based generative models for downstream tasks via PyTorch frameworks and HuggingFace, pillow, and OpenCV. Generative model evaluation of text descriptions and image labels, using accuracy and semantic similarity.
- Designed the research to analyze online visual and textual narrative representations of the anti-femicide movement in Mexico using network analysis, computational social science, and computer vision / natural language processing (neural network) applications. Implement simulation and testSing on multimodal LLM classification and generative captioning using Llama 3 Vision Qwen2.5 VL, BLIP-2 to to effectively test and evaluate LLM performance across various image classification tasks. The open science AI project goal: understand fragmented news on social media and evaluate semantic relationships between image posts.
- Applied graphical analysis on image and text embeddings using Variational Autoencoders (VAE) within a Graphical Attention Network (GAT) model to compare classifications of the collected annotated image data to other benchmark graph datasets, Cora, flicker. Evaluated all VAE embeddings using HDBSCAN and other evaluation metrics on all image data for unsupervised learning support.

Data Science Fellow | *Laboratory for Analytic Sciences* - Summer Conference of Applied Data Science; North Carolina State University, Raleigh, North Carolina, 06/2024 – 07/2024

- **Compute and Processing Resources Awarded:** Unlimited OpenAI tokens through AWS Commander allocation. Unlimited access to g1-g4 HPC instances. 3+ TB of storage. **Awarded \$11,500 Stipend.**
- NLP researcher comparing chunk-size methods for text summarization. Used extractive summarization occams and abstractive summarization, Retrieval Augmentation Generation (RAG) to test different text segmentation methods. Used LangChain architecture and created a vector embedding database for evaluation RAGAs(LLM RAG evaluator). Used 2024 TREC Challenge data-MSMarco docs and QA for analysis.
- Project lead for computer vision comparison research of video object tracking models: zero-shot transformers (OWL-ViT, GPT-4o, Phi3) on noisy video surveillance data. The goal was to explore improvements for tracking object obfuscation and detecting unknown/unconventional objects in the noisy video data. Found that pre-trained image models provided more accurate structured outputs.

Graduate Teaching Assistant | *University of Arizona College of Information* - Tucson, AZ, Fall 2023 – Spring 2024

- INFO 523: Data Mining and Discovery; INFO 523: Data Analysis and Visualization.

Roots for Resilience Data Science Fellowship | *University of Arizona Data Science Institute*, 08/2023 – 01/2024

- **Compute and Processing Resources Awarded:** 3 TB of data storage; 20,000 compute units/year; Ability to run 4 concurrent jobs with instance allocation. **\$6,000 Grant**
- **Data Management:** Created and maintained a Data Management Plan for a social media project, using Open Science Framework in collaboration with CyVerse. DataOps pipelines, Version Control, GitHub, and Reproducibility, Docker.

EXPERIENCE & RESEARCH (Continued)

- Used SQL queries and database management to connect MySQL database to leaflet map built in JavaScript using Python library, flask API, Interactive Leaflet map creation for social movement group in Mexico Ellastienennombre.org. Created search feature to retrieve data points on map that represent femicide victims in Ciudad Juarez, Mexico.

LEADING Data Science Fellowship 2023 Cohort | *University of Rochester Remote, 06/2023 – 01/2024*

- Led the creation of an Open Source Data Pipeline to collect and analyze bibliometric data by University of Rochester Researchers to identify their publication behavior. Collected, cleaned, and analyzed 500K+ dataset metadata using APIs from various repositories. Set-up data for analysis and reproducibility.
- Used trend and network analysis methods including regression (decision trees), time series analysis, structural analysis, and prediction analysis for network validation of researcher behavior.

Data Science Grad Assistant | *University Analytics & Institutional Research, Tucson, AZ 08/2021 – 08/2023*

- *NLP Sentiment Analysis:* Analyze 4M+ student course survey responses using Natural language processing to identify sentiment, word frequencies and word synonyms for curriculum in various departments on campus. Increase faculty participation in curriculum review across schools on campus.
- *Machine Learning* - Spearheaded the Women in Science and Engineering Retention report project which uses survival and churn analysis. These analyses review churn or dropout rates of 37K+ students over ten years. The analysis incorporates machine learning applications including decision tree and random forest models along with other time-based statistical methods. I identified student financial, cultural, academic, and demographic characteristics that either debilitate or foster success towards graduating college.
- *Dashboard Creation* - Using automated and qualitative approaches, I developed End-to-end application of tuition and headcount dashboard visualizations and analysis using R and SQL. Created aggregated data tables from an internal Oracle data warehouse using descriptive statistics and inflation information of campus-wide Net Tuition Revenue (NTR). The dashboard is currently used by decision makers on campus.

Computational Tools and Thinking - Teaching Fellow | *University of Texas Austin, 06/2022 - 7/2022*

- Developed ten, two-hour lesson plans for a computational tool and thinking survey course that covered R, Python, Social Network Analysis, Design Theory, and coding best practices. Taught to a wide range of experience levels that included 25 high achieving undergraduate students from STEM and Social Sciences.
- Held office hours to support outside research and enabled the development of a 20-month research design process. Participated in community building activities with other distinguished fellows as well as with our students. Maintained mentor level contact with students in coding and research analysis applications.

Data Science Ambassador | *University Arizona Data Science Institute, Tucson 08/2021 – 08/2022*

- Data Carpentries Instructor as of 2/14/2022 -Live coding instructor and helper for Data Analysis and Visualization with R for Social Scientists, Introduction to Python, Open Refine, and other data science skills.
- Project Manager: data science community support and resources for the CSS certificate and iSchool. Created and Organized Projects in Progress. Fostered a Data Science community environment for researchers on campus to collaborate and share projects ideas and progress. Helped consult 3 - 5 projects over the span of two semesters (over 20 researchers) using data science methods monthly.
- Helped organize Stanford-based Women in Data Science satellite conference, organized and moderated the Computational Social Science panel. The April 2022 event had around 400 participants on/offline.

Data Analyst Research Assistant | *Arizona Medical Education Research Initiative, UA, Tucson, 01/2021 – 08/2021*

- *Multivariate Statistical Analysis:* Applied statistical analysis an inference to collected data to review and test the University of Arizona's Medical School's performance for future accreditation. Types of data include survey data, archived data, and collected government data from large data systems. Used both R and Python to apply analysis of variance, chi-square tests, validation checks, regression, and correlation.

EXPERIENCE & RESEARCH (Continued)

- *Portfolio Analysis and Program Use*: Worked with Power BI, Excel, R and Python to clean, manage, transform, analyze, iterate, and visualize data. Created data-driven program evaluation reports for various research questions about the medical education program using statistical analysis and visualization.

Survey Data Manager | Minerva Project: University of Arizona, Tucson, AZ, Summer 2021

- NSF Grant funded project: Contributed to maintenance of ETL pipeline for 8k+ unique and personalized surveys in English, Spanish, Pashto and Dari. Applied data cleaning, wrangling, and statistical analysis to a databank of crowd-sourced history of armed conflict in Afghanistan and Colombia for the US DoD.
- Troubleshooted and applied forensic analysis to problem solve issues in the pipeline and iterate over processes for optimization. Automated forecasting of dynamic data. Completed time series visualizations for dynamic datasets by manipulating data in R to show the territorial control of Criminal Organizations.

Community Outreach and Program Coordinator | Lutheran Social Services of the Southwest K-12 Refugee Education (AmeriCorps State Member), Tucson, AZ, 08/2016 – 08/2019

- Arizona Serve State Member working with the non-profit organization, LSS – K-12 Refugee Education CENTER, by providing academic support, empowerment, and leadership skills to refugee students in Tucson, Arizona. Provided assessment and outreach for Tucson schools with large refugee students by developing a database to track community resources and partnerships. Connected students to local academic advancement resources.
- Created a sustainable measurement process to identify the impact of LSS-SW Education services based on student survey and school-based data. Produced a report of the year's work, updated Refugee Youth Services Handbook, for stakeholders. Kept LSS-SW a primary education resource for Tucson refugees.

Digital Content Specialist | Tetakawi, Tucson, AZ, 08/2016 – 05/2019

- Conducted field research on the Mexican labor market for advisory services branch. Provide report editing, and data check to research findings. Researched industry resources on economic, regulatory and policy events in Mexico and U.S. trade relations on company website.

Web Designer & Content Management | Georgetown University, Washington, D.C. 2015 - 2016

- Applied design thinking consulting to help the marketing, IT, and communications director transition to a new website platform. Solved Problems with coding elements using bootstrap and Drupal platforms. Applied UX design methodology and analysis. <https://mccourt.georgetown.edu/>. Applied quality assurance review and web design best practices for multiple website projects.

LEADERSHIP

PhD Community Manager (University of Arizona iSchool), 2021 – Present: work across iSchool disciplines to lead and coordinate strategic programs and opportunities that advance research, innovation, and community outreach.

Graduate Student Representative on Search Committee for Dean of the College of Information Science, 2025 - 2026

AI Steering Committee | Graduate Student Representative, University of Arizona, 2023 - 2026

NLP Reading Group UofA, 2019-Present

ASIS&T SIG – Visualization, Sound, and Image: Secretary Chair 2021-2022

WiDS-Tucson (Women in Data Science), Panel Sessions Organizer, Data Blitz Committees 2021 -2022

RezBaz - Presenter & Moderator, Network Analysis 2021; R Visualization 2021 and 2022

iSchool Colloquium, Organizer: Moderated Lectures, led school-wide Faculty Research Blitz 2020 – 2021

Community Outreach and Program Coordinator | Lutheran Social Services of the Southwest K-12 Refugee Education (AmeriCorps State Member), Tucson, AZ, 08/2016 – 08/2019

COURSEWORK AND CERTIFICATIONS

Advanced Natural Language Processing – Computer Science, Fall 2024
Introduction to Computer Vision – Computer Science, Fall 2023
Introduction to Artificial Intelligence– College of Information, Spring 2022
Bias in Information Retrieval Systems (Graduate Seminar) – College of Information, Spring 2022
Probabilistic Graphical Models, Computer Science – Spring 2022
Theory of Probability – School of Statistics, Fall 2021
Visual Epistemologies of Media and Technology (Graduate Seminar) - College of Information, Spring 2021
Applied Social Network Analysis– College of Information, Spring 2021
Social Movements and Collective Action – School of Sociology, Fall 2020
Neural Networks– College of Information, Fall 2020
Bayesian Models and Inference– College of Information, Fall 2020
Human Rights and Technology – College of Information, Fall 2020
Intro to Machine Learning- Electrical and Computer Engineering, Spring 2020
Micro-dynamics of Political and Organized Criminal Violence – Government and Public Policy, Spring 2020
Presenting Data and Information- Edward Tufte, 2016